

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay_compression.
- Best TSPLIB holdout gap: tsplib_failure_replay_compression.
- Best synthetic holdout gap: tsplib_random_replay.

Run Metadata

- run_name: run_20260513_141159_j
- started_at_local: 2026-05-13 14:11:59
- finished_at_local: 2026-05-13 14:22:30
- duration_hhmm: 00:11
- duration_seconds: 630.69
- seed_offset: 9000
- replicate_label: j
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.887401	0.56	-0.028065
tsplib_random_replay	random	score_only	False	0.147693	0.0	0.093986	0.701982	0.76	0.107141
tsplib_failure_replay	failure	score_only	False	0.197828	0.0	0.125891	0.0	0.76	0.0
tsplib_failure_replay_compression	failure	novelty_gate	True	0.119024	0.0	0.075743	0.735647	0.76	0.020953

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.887401, and final complexity 0.56.
- Adaptation efficiency -0.028065 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.147693, synthetic 0.0, combined 0.093986.
- Accepted-epoch count 2, mean accepted code novelty 0.701982, and final complexity 0.76.
- Adaptation efficiency 0.107141 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.147693 across 7 instances; family means: ch=0.119814, kroD=0.198366, pcb=0.191677, pr=0.248144, rd=0.12048, st=0.035556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3274, "family": "a", "name": "a280", "optimality_gap": 0.269484}, {"best_known_cost": 14379, "cost": 17172, "family": "lin", "name": "lin105", "optimality_gap": 0.194242}, {"best_known_cost": 629, "cost": 737, "family": "eil", "name": "eil101", "optimality_gap": 0.171701}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.197828, synthetic 0.0, combined 0.125891.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.197828 across 7 instances; family means: ch=0.260217, kroD=0.216446, pcb=0.189413, pr=0.242939, rd=0.18445, st=0.031111.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3418, "family": "a", "name": "a280", "optimality_gap": 0.32532}, {"best_known_cost": 629, "cost": 771, "family": "eil", "name": "eil101", "optimality_gap": 0.225755}, {"best_known_cost": 7542, "cost": 9166, "family": "berlin", "name": "berlin52", "optimality_gap": 0.215327}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.119024, synthetic 0.0, combined 0.075743.
- Accepted-epoch count 4, mean accepted code novelty 0.735647, and final complexity 0.76.

- Adaptation efficiency 0.020953 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 4.
- Panel heldout_tsplib mean gap 0.119024 across 7 instances; family means: ch=0.090423, kroD=0.122617, pcb=0.218461, pr=0.116107, rd=0.066245, st=0.128889.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3516, "family": "a", "name": "a280", "optimality_gap": 0.363319}, {"best_known_cost": 629, "cost": 778, "family": "eil", "name": "eil101", "optimality_gap": 0.236884}, {"best_known_cost": 7542, "cost": 9087, "family": "berlin", "name": "berlin52", "optimality_gap": 0.204853}]

Judge Appendix

Summary of Results for Replay-Aware TSP Benchmark

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Replay Mode	Code Novelty (mean)	Complexity	Notes on Transfer and Novelty
tsplib_failure_replay_compression	**0.1190**	0.0	**0.0757**	failure + compression	0.7356	0.76	Best transfer (TSPLIB and synthetic heldout); code novelty lower than no replay but transfer substantially improved; compression-aware replay aids transfer
tsplib_random_replay	0.1477	**0.0**	0.09399	random replay	0.7019	0.76	Best synthetic holdout; better transfer gap than no replay and failure replay; code novelty lower than no replay but improved transfer
tsplib_failure_replay	0.1978	0.0	0.1259	failure replay	0.0	0.76	No code novelty; worse transfer than random replay and failure replay with compression
tsplib_no_replay	0.2134	0.0649	0.1594	no replay	**0.8874**	0.56	Highest code novelty but worst transfer metrics; replay mechanisms improve transfer despite drop in novelty

Conservative Interpretation

- ****Transfer Evidence (Held-out TSPLIB and Synthetic Gaps):****
- The ****tsplib_failure_replay_compression**** condition yields the lowest TSPLIB and transfer gaps (0.1190 and 0.0757), indicating superior generalization and transfer to held-out benchmark instances.
- The ****tsplib_random_replay**** condition also shows strong transfer performance (TSPLIB gap 0.1477, transfer gap 0.09399) and is best in synthetic holdouts (mean synthetic gap 0.0).
- Both failure replay alone and no replay yield higher transfer gaps, with no replay showing the worst transfer (TSPLIB gap 0.2134, transfer gap 0.1594).
- ****Code Novelty vs Transfer:****
- The no replay condition exhibits the highest code novelty (~0.887), yet has the poorest transfer metrics.

- All replay modes reduce code novelty (0 to ~0.73 range) while improving transfer gaps; notably, **failure replay + compression** achieves the best transfer with moderate code novelty.
- This demonstrates a clear pattern: **code novelty decreases while transfer performance improves**, implying lexical novelty does not directly equate to algorithmic invention or better transfer.
- **Replay Method Distinctions:**
- **No Replay:** Highest code novelty but worst generalization and transfer.
- **Random Replay:** Good transfer, some code novelty retained, best synthetic gap.
- **Failure Replay:** No code novelty retained, transfer worse than compression-aware failure replay.
- **Failure Replay + Compression:** Best transfer performance, moderate code novelty, higher complexity (0.76), and specialized final behavior ("clustered_constructor").
- **Algorithmic Behavior & Complexity:**
- Replay conditions (especially compression-aware failure replay) maintain or increase complexity (0.76) compared to no replay (0.56), suggesting more sophisticated/structured behavior contributing to transfer improvements.
- Final behavior profiles differ: "balanced" for no replay/random/failure replay, "clustered_constructor" for failure replay with compression.

Conclusion

- **Best transfer results are obtained with failure replay combined with compression pressure,** outperforming both random and vanilla failure replay, and no replay conditions on both held-out TSPLIB and synthetic benchmarks.
- **This improvement in transfer comes at the cost of reduced code novelty**, showing that replay affects algorithmic generalization more than it spurs lexical novelty.
- **Random replay is a viable alternative yielding good transfer and synthetic performance with moderate novelty.**
- **No replay affords high novelty but at notable transfer and performance costs.**
- Replay mechanisms, especially compression-aware failure replay, effectively guide algorithms toward improved transfer metrics, supported by deterministic optimality gaps and not narrative speculation.

Recommendation for Benchmark Usage

Use **tsplib_failure_replay_compression** as the primary benchmark condition for evaluating transfer-aware TSP algorithms due to its demonstrably superior transfer performance. Consider **tsplib_random_replay** as an alternative with competitive transfer and zero synthetic gaps. Avoid no replay if transfer is the evaluation priority despite its higher code novelty.